

PopGenAgent: An Agent-Mediated Workflow for Population Genetics Analyses

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Abstract

Population genetics analyses typically require orchestrating multiple tools across heterogeneous tasks, from data curation to inference and visualization. In practice, a central bottleneck lies not only in reproducibility, but in maintaining consistency between intermediate diagnostics, analytical decisions, and downstream results as analyses evolve. Here we present PopGenAgent, an agentic workflow that formalizes population-genetic analyses as explicit, multi-step plans with declared inputs, outputs, and dependencies. Each step is instantiated from curated tool and visualization templates, while all intermediate artefacts are preserved within a unified, inspectable context. An interactive agent interface supports iterative plan construction and revision, and leverages accumulated outputs to assist interpretation and report drafting. In a case study of 26 populations from the 1000 Genomes Project, starting from a filtered PLINK dataset, PopGenAgent coordinated analyses including ROH, LD decay, PCA, ADMIXTURE, TreeMix, and f_3 . By structuring analyses as explicit, revisable plans with preserved intermediate states, PopGenAgent provides a transparent and extensible environment for conducting population-genetic analyses.

Keywords Population Genetics, Workflow, Pipeline Bioinformatics Agent

Introduction

Population genetics studies combine multiple analyses to characterize genetic diversity and population structure Hamilton (2021). In practice, however, the main challenge is often not running an individual program, but keeping file transformations, parameter choices, intermediate diagnostics, and downstream biological interpretation coherent as an analysis is revised. A typical study may move repeatedly across genotype-format conversions, pruning decisions, structure analyses, diversity summaries, and admixture diagnostics before results are ready for reporting.

Reproducible workflow systems alleviate part of this burden by standardizing software environments and execution patterns Wratten et al. (2021). General frameworks such as nf-core Ewels et al. (2020), GenPipes Bourgey et al. (2019) provide reusable pipelines for common analyses, while population genetics specific platforms extend this paradigm to domain workflows: for example, the Pop-Gen Pipeline Platform (PPP) structures VCF processing and downstream utilities Webb et al. (2021), and PAPipe integrates whole-genome processing with predefined population-genetic analyses Park et al. (2024). Despite these advances, a substantial portion of analytical effort remains outside the formalized pipeline structure, arising in the iterative stages where users inspect intermediate outputs, adjust filtering or pruning decisions, and ensure that downstream analyses and summaries remain consistent with those evolving choices Kopelman et al. (2015); Li and Liu (2018).

This issue is particularly acute in population genetics because interpretation usually depends on comparing several complementary outputs rather than on a single endpoint. For instance, analysts often examine heterozygosity and ROH summaries, LD decay, PCA, model-based ancestry inference, and admixture diagnostics together, while also managing conversions among tool-specific formats and plot-ready inputs Kamvar et al. (2015); Leigh et al. (2015). When an upstream choice changes, the practical challenge is not only rerunning commands, but also maintaining a clear record of what changed, which outputs should be regenerated, and how revised results should propagate into figures and narrative text.

Large language models offer a complementary interface for reducing scripting burden and supporting interactive analysis Zhao et al. (2023); Chang et al. (2024); Kasneci et al. (2023). In computational biology, early LLM-based systems have demonstrated tool use and workflow assistance in domains such as genome analysis and transcriptomics Zhou et al. (2024); Huang et al. (2025); Su et al. (2025). For population genetics, however, the main need is not open-ended code generation. It is reliable coordination of standard analyses under realistic failure modes, together with publication-ready outputs that remain traceable to the intermediate artefacts from which they were derived. An effective system therefore needs explicit workflow state, inspectable execution steps, and support for stepwise revision rather than one-shot script generation alone.

PopGenAgent was developed to support this mode of analysis by providing a workflow framework that maintains analysis plans, execution steps, intermediate artefacts, and reportable

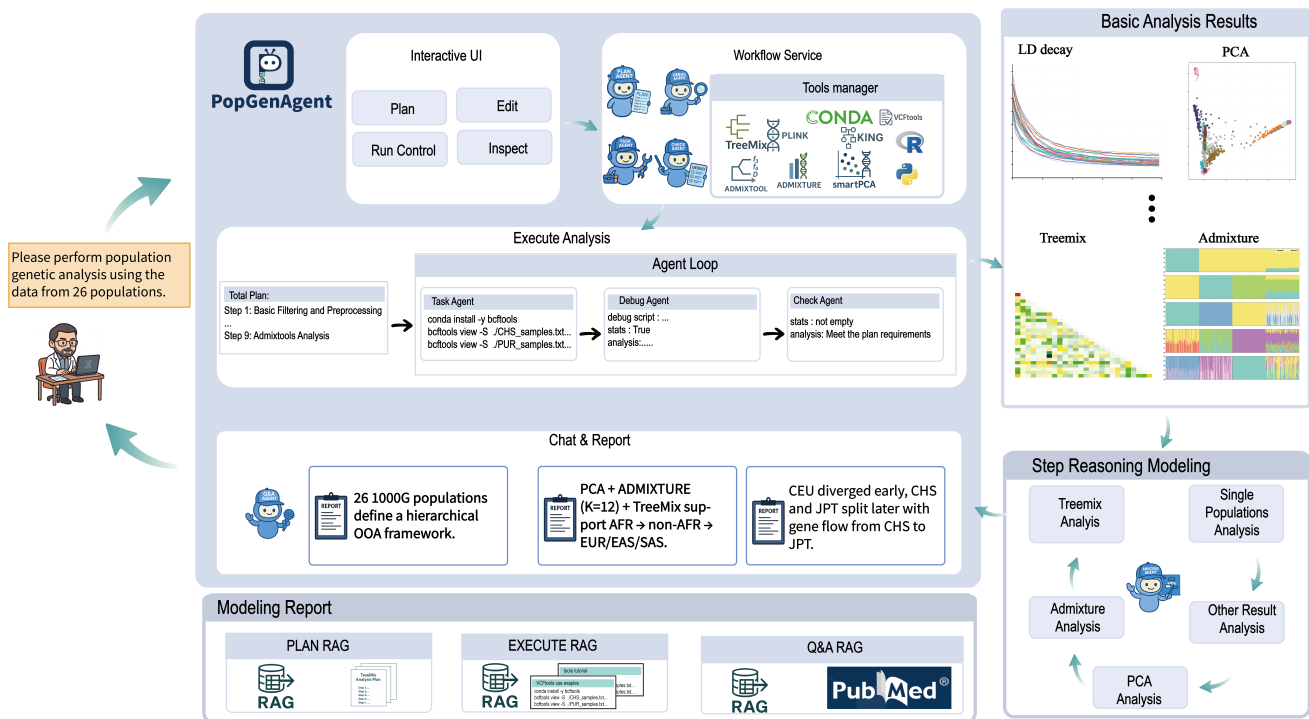


Figure 1 Overview of PopGenAgent. PopGenAgent uses an agent-mediated interface to translate an analysis goal and genotype dataset into an explicit multi-step plan assembled from curated templates. Steps are executed via external analysis tools and plotting scripts, while intermediate and final artefacts are organized for inspection and revision within the same analysis context. The agent can also provide workflow-oriented assistance, including, when configured, literature-backed Q&A and Markdown report drafting grounded in the saved outputs.

outputs within a unified context. Analyses begin from prepared genotype datasets and are structured as explicit multi-step plans with declared inputs and expected outputs, instantiated from curated tool and visualization templates. An agent-mediated interface supports iterative plan construction and revision, while leveraging accumulated outputs to assist interpretation and reporting. In the following, we describe the analytical coverage and system design, and illustrate the workflow through a 26-population case study from the 1000 Genomes Project.

Results

PopGenAgent represents multi-step analyses as explicit workflows

PopGenAgent separates analytical reasoning from code generation by constraining the agent to a curated set of implemented analysis tools and plotting scripts. Rather than relying on the LLM to generate analysis code dynamically from its own knowledge, PopGenAgent enables the agent to assemble workflows by selecting and parameterizing predefined computational and visualization components. This design improves reliability and analytical consistency by reducing uncontrolled code synthesis, while allowing analyses, intermediate artefacts, and final outputs to remain organized within a persistent workflow context (Fig. 1).

PopGenAgent covers tools spanning data preparation and quality control, within-population diversity summaries, population-structure analyses, etc. (Table 1). In the interface,

these analyses are represented by plans with declared inputs and expected outputs (Supplementary Fig. S1) and supported by stored tool references together with reusable task and visualization templates (Supplementary Fig. S2). The coverage summarized in Table 1 corresponds to the workflows exercised in the 1000 Genomes example below.

Explicit plans and stored artefacts support intermediate-state review

A core feature of PopGenAgent is that workflow intent remains visible throughout an analysis. The saved plan lists step descriptions, required inputs, and expected outputs (Supplementary Fig. S1), making it possible to review what will be executed and which artefacts should be produced before later steps are run. Intermediate plots and tables can then be examined alongside the plan that generated them, and upstream steps can be revised before downstream structure or admixture summaries are regenerated.

To support consistent execution across analyses, PopGenAgent maintains stored tool references and reusable task and visualization templates (Supplementary Fig. S2). These resources capture command patterns, configurable parameters, and expected file organization for supported workflows. Because planning artefacts and generated outputs are preserved within the same analysis context, the agent can also provide output-grounded assistance and assemble report drafts that link narrative text to the saved figures and tables (Fig. 1).

Module	Representative tasks and outputs	Representative tools
Data preparation and QC	Genotype/VCF filtering, LD pruning, and relatedness screening for downstream structure and admixture analyses	PLINK, VCFtools, KING
Diversity and LD summaries	ROH burden and length composition, observed/expected heterozygosity, inbreeding coefficients, and LD decay summaries (Fig. 2)	PLINK
Population structure	PCA projections and variance summaries, ADMIXTURE cross-validation scans, and ancestry bar plots (Fig. 3; Supplementary Figs. S3 and S5)	smartPCA, ADMIXTURE
Admixture and tree-based method	TreeMix model scans, fitted graphs, and residual diagnostics (Fig. 3; Supplementary Fig. S4) together with allele-sharing summaries such as representative f_3 panels (Fig. 2)	TreeMix, ADMIXTOOLS

Table 1. Representative analytical coverage implemented in PopGenAgent and illustrated in this manuscript. The table summarizes core modules, outputs, and external tools coordinated by PopGenAgent.

A 1000 Genomes example illustrates routine downstream outputs from population-genetic analyses

We analyzed the 26-population panel from the 1000 Genomes Project Auton et al. (2015) to illustrate a routine set of downstream outputs generated by PopGenAgent. The analysis started from a filtered PLINK dataset, and derivative files such as LD-pruned inputs and EIGENSTRAT conversions were generated as required by downstream methods. Figures 2 and 3 show representative outputs from this analysis, and Supplementary Figs. S3–S5 provide the corresponding model-scan panels and supporting summaries, including the full ADMIXTURE K scan, additional TreeMix fits, and the PCA scree plot.

Figure 2 summarizes within-population diversity, autozygosity, and representative allele-sharing output. Panels a and b show observed heterozygosity across populations and its comparison with expected values, and panel c shows the corresponding inbreeding coefficient distributions. Panel d summarizes ROH burden across populations in terms of total ROH length per individual and ROH segment counts. Panel e shows population-specific LD decay curves, and panel f reports representative f_3 statistics with standard errors for selected population triplets.

Figure 3 summarizes complementary structure and graph-based outputs. PCA separates major continental groupings and reveals gradients among populations (Fig. 3a), while a descriptive dendrogram computed from population means of the first five PCs provides a compact population-level summary (Fig. 3d; Methods). ADMIXTURE cross-validation errors are tracked across the tested range of K values (Fig. 3c), and Supplementary Fig. S3 shows the corresponding ancestry bar plots. We display the $K = 5$ solution in the main text as a representative continental-scale summary (Fig. 3g).

A series of TreeMix models were generated within the same workflow. Figure 3 shows log-likelihood values across alternative numbers of migration edges (Fig. 3b), residual covariance patterns (Fig. 3e), and fitted graphs (Fig. 3f), while Supplementary Fig. S4 shows additional TreeMix fits and Supplementary Fig. S5 reports the PCA scree plot. Together, these outputs illustrate that PopGenAgent can coordinate Population genetics analysis tools and results within one inspectable workflow context.

Discussion

Population-genetic analyses are often framed as pipelines, but in practice rely on iterative inspection and revision, where intermediate diagnostics must be kept consistent with downstream results. PopGenAgent addresses this gap by organizing analyses as explicit, revisable plans with preserved intermediate artefacts in a unified context. A key design choice is to constrain execution to curated tools and visualization templates, rather than relying on dynamically generated code, improving consistency and reducing variability across analyses. Compared with existing workflow systems that standardize execution, PopGenAgent focuses on structuring the analysis process itself and maintaining alignment between analytical steps and outputs. Limitations include dependence on predefined components and limited coverage of specialized analyses. Future work will expand supported tools and improve flexibility for more complex workflows.

Methods

System Architecture and Persisted Analysis Context

PopGenAgent is a web-based workflow system for Population genetics analyses. In the current implementation, an LLM-based agent serves as a control layer that translates user goals into explicit multi-step plans, retrieves relevant workflow resources, proposes plan revisions in response to intermediate outputs, and assists interpretation and documentation within the current analysis context. Analytical execution is anchored to curated tool and visualization templates rather than free-form tool calling: each step declares required inputs and expected outputs and is instantiated as an inspectable script that calls the underlying analysis program or plotting routine.

Each analysis context is persisted on disk, including the plan, instantiated step scripts, execution logs, and an indexed manifest of generated figures and tables. This persisted state allows users to inspect intermediate artefacts, revise earlier steps, and draft reports without losing the provenance of downstream outputs.

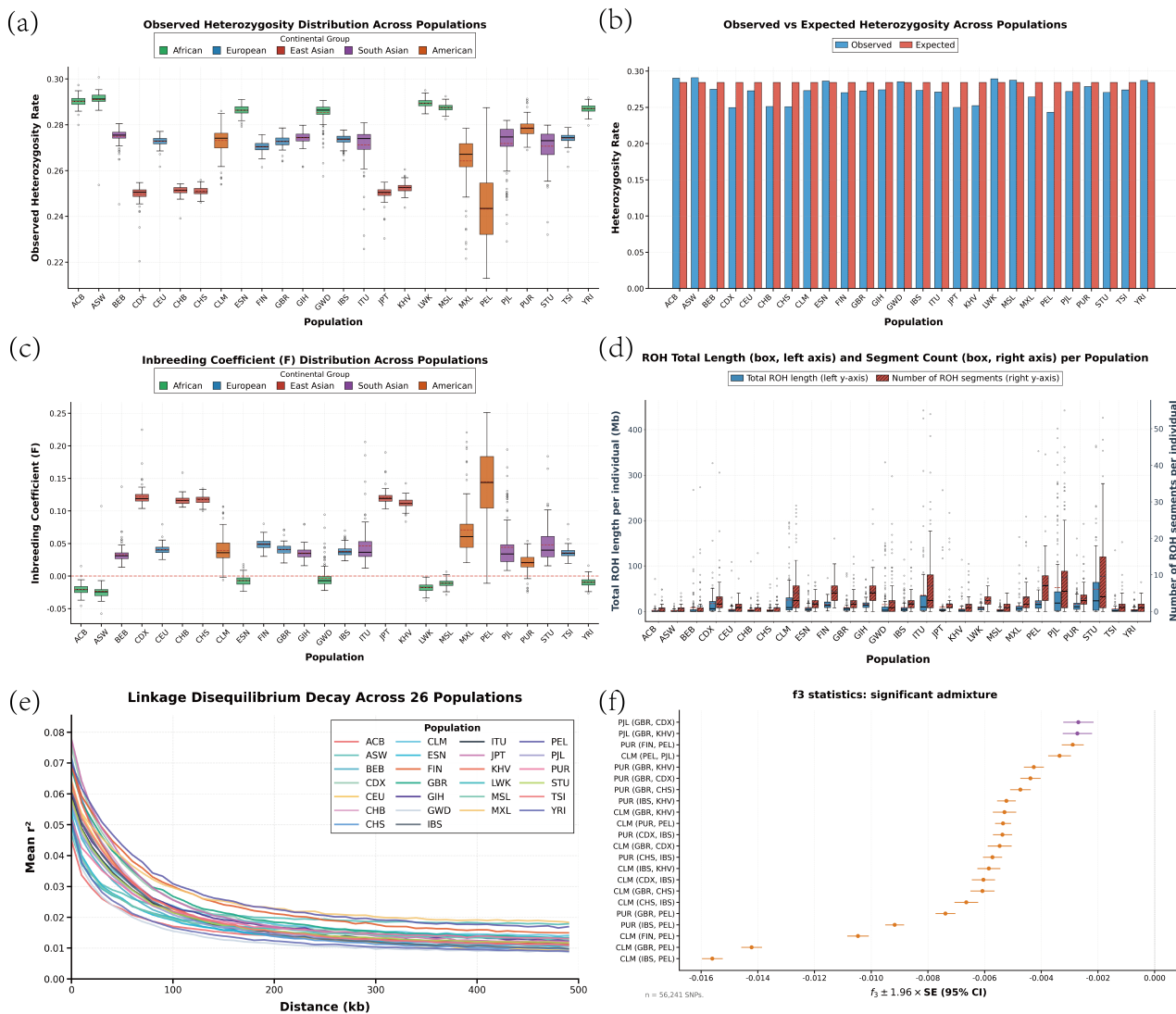


Figure 2 Summary statistics from the 1000 Genomes example analysis. (a) Observed heterozygosity distribution across populations. (b) Observed versus expected heterozygosity across populations. (c) Inbreeding coefficient (F) distribution across populations. (d) Distribution of total ROH length and ROH segment count across populations. (e) Genome-wide LD decay curves across populations. (f) Representative f_3 statistics for selected population triplets.

Workflow Specification Through Plans and Templates

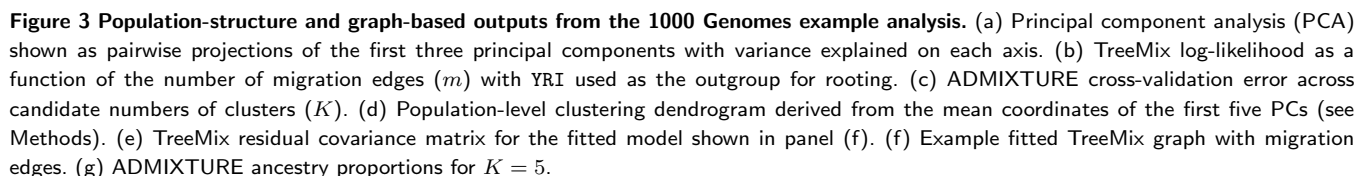
Workflow plans are structured representations of an analysis in which each step is associated with a description, declared input files, and expected outputs (Supplementary Fig. S1). Step templates encode tool-specific command patterns, configurable parameters, and file-organization conventions, whereas visualization templates convert raw outputs into standardized plots and summary panels (Supplementary Fig. S2). Together, the plan and template layers define what the workflow is expected to produce and provide inspectable specifications that can be revised before later steps are executed.

Stepwise Execution and Output Validation

Execution proceeds stepwise through the saved plan. After each step, PopGenAgent checks whether the declared outputs are present and satisfy basic integrity criteria, such as non-empty files and expected formats. If a step fails, the system retains diagnostic information for inspection and, when the failure matches a supported template-level adjustment, may attempt a limited retry. This orchestration model is adapted from the framework introduced in BioMaster Su et al. (2025) and specialized here for Population genetics workflows.

Implemented Analytical Workflows

The current template library coordinates downstream preprocessing and quality-control tasks (PLINK Chang et al. (2015), VCFtools Danecek et al. (2011),



KING Manichaikul et al. (2010)), population-structure analyses (smartPCA/EIGENSOFT Patterson et al. (2006), ADMIXTURE Alexander et al. (2009)), graph-based modeling (TreeMix Pickrell and Pritchard (2012)), and allele-sharing statistics in the f -statistics framework (ADMIXTOOLS Patterson et al. (2012)). In the 1000 Genomes example, PopGenAgent starts from a filtered genotype dataset in PLINK format; derived files such as LD-pruned SNP sets and EIGENSTRAT conversions are generated as required by downstream methods.

In addition to tool execution, plotting templates generate standardized panels for ROH and heterozygosity summaries, LD decay curves, PCA projections and scree plots, ADMIXTURE cross-validation curves and ancestry bar plots, and TreeMix likelihood and residual diagnostics (Figures 2–3; Supplementary Figs. S3–S5). For a descriptive population-level summary, we compute Euclidean distances among population means of the first five PCA coordinates and visualize Ward hierarchical clustering as a dendrogram. This visualization is used as a compact summary of population-level separation in PCA space and is not intended as phylogenetic inference.

Output-Grounded Assistance and Report Generation

Within the current analysis context, PopGenAgent can answer workflow-oriented questions by drawing on recent dialogue, stored planning information, and a curated knowledge store; when configured, it can additionally retrieve relevant PubMed abstracts for background questions. For documentation, the system assembles a Markdown report draft by combining the saved plan with the indexed figures and tables generated during execution, providing a starting point for an analysis write-up grounded in the produced artefacts. Drafted text should be checked against the saved outputs before reuse.

Data and Code Availability

The analyses reported here use publicly released data from the 1000 Genomes Project, maintained by the International Genome Sample Resource (IGSR). Sequence reads, alignments, and phased variant calls for the project can be accessed through the IGSR portal: internationalgenome.org. An archived release associated with this manuscript is available at zenodo.org/records/18919671.

The PopGenAgent source code is publicly available at github.com/ai4nucleome/POPGENAGENT. The public repository includes an `environment.yml` file for software setup. Supplementary figures are provided in a separate Supplementary Information file.

Competing interests

The authors declare no competing interests.

Generative AI statement.

During the preparation of this work, the authors used ChatGPT to assist with language editing and manuscript revision. After using this tool or service, the authors reviewed and edited the

content as needed and take full responsibility for the content of the publication.

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